Aqiba Abdul Qadir

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Achievements

Academic Excellence — Summa Cum Laude and Gold Medalist
IELTS — Achieved an overall score of 8.0 bands, demonstrating C1 level English proficiency.
NSCT— Scored 82/100 (99.8th percentile), ranking among top 0.2% of 40,784 IT students in Pakistan.
Best FYDP Award — Recognised for best Final Year Project at FYDP Exhibition held at CIS department.
Hackathon Winner — Secured 3 victories in hackathons, including a solo first-place win.
International Competitions — Participated in 15+ national and international competitions including Meta HackerCup, Advent of Code, CodeForces, UC Berkeley's CALICO, etc.
Competitive Programming — Solved 1000+ LeetCode problems with a contest rating of 1620+, placing among the top 100 in country-wide rankings.

Education

Bachelor of Computer Systems Engineering Sep 2022 – Jul 2026
NED University of Engineering & Technology Karachi, Pakistan

- **CGPA:** 3.991/4.0
- **Relevant Coursework:** OS, DSA, AI, ML, Digital System Design, Computer Organisation and Architecture, Digital Signal Processing, Parallel Programming
- **Final Year Project (Ongoing):**
  FPGA-based Hardware Acceleration of Post-Quantum Cryptography
  An FPGA-based system for CRYSTALS-Dilithium acceleration addressing post-quantum cryptographic performance bottlenecks to enable quantum-secure digital signatures.
  Technologies: C, Verilog, Artix-V FPGA, Vivado Design Suite

Professional Experience

Associate Data Scientist July 2026 – Present
Veeam Software Karachi, Pakistan

- Integrating Data Science models into Securiti AI's content classification pipeline, supporting model evaluation and performance optimization.
- Enhancing classification through rule-based feature engineering and fine-tuned LoRA adapters for specialized information extraction.

Data Science Summer Intern Jul 2025 – Aug 2025
Securiti.ai Karachi, Pakistan

- Optimised a document classification system for the securiti product developing a single LlaMa model to replace a complex, hybrid solution.
- Improved model accuracy over 37 document types up to 96.5%.

Personal Projects

PII Masking with Named Entity Recognition
NLP system to detect and mask Personally Identifiable Information (PII)

- Developed a high-accuracy (99.56%) PII detection and redaction system by fine-tuning a transformer model and creating a synthetic data pipeline to solve training data limitations.
- **Technologies:** Python, Transformers, Google Colab

Row Buffer Architecture for Neighbourhood Image Processing
Research Paper Implementation

- Achieved upto 87.5% BRAM savings compared to conventional row buffers
- Retrieves multiple pixels per clock cycle with 1 clock/pixel efficiency

- Modular design and test benches to ensure design correctness
- **Technologies:** Verilog, Icarus, GTKWave

Hardware Acceleration of Schnorr's Signature Scheme
Discrete Logarithmic Hardness Public-Key Cryptographic Algorithm
- Developed an FPGA-based acceleration for Schnorr's digital signature scheme
- Parameterised the design for different key lengths improving design flexibility
- Validated cryptographic modules with 100% functional accuracy
- **Technologies:** Verilog, Vivado, Artix-7 FPGA

Computational Visualization Projects
Interactive Complex Concept Visualizers
- **DNA Sequence Alignment Visualizer:** Visualised Needleman-Wunsch and Smith-Waterman algorithms
- **Quantum Playground:** Developed an interactive web app to visualize quantum concepts (entanglement, superposition, teleportation, Bell states) with a circuit builder and Bloch sphere visualizations.
- **Technologies:** Python, NumPy, Matplotlib, Qiskit, Streamlit

Hackathon Projects

QubitCoders
LeetCode Profile Analyzer | PakAngels GenAI Hackathon | Top 10 projects
- Developed an AI-powered web application that analyzes a user's LeetCode profile to classify their programming expertise using Fuzzy Logic algorithms.
- Integrated the Gemini API to generate personalized, context-aware coding challenge recommendations for skill improvement.
- **Technologies:** Python, Gemini API, Fuzzy Logic, VS Code, Streamlit, GitHub

Diabetic Retinopathy Detector
AI-Powered Medical Screening Tool | Infyma Hackathon | Runner-up
- Optimised a CNN, fine-tuned for high-accuracy multi-class classification of retinal disease severity.
- Built an intuitive web interface with Streamlit to allow users to easily upload images and receive instant, explainable predictions.
- **Technologies:** Python, TensorFlow, Transfer Learning, Streamlit, Computer Vision, EfficientNet B3

Volunteer and Teaching Experience

CIS Community, NEDUET Jan 2025 – Aug 2025
Technical and Innovation Team Lead Karachi, Pakistan
- Cultivating a university-wide problem-solving environment by organizing and leading coding and DSA activities helping 150+ students improve their skills.
- Driving a new initiative to encourage team members to develop innovative, revenue-generating projects that enhance their skill sets.

Code In Place'24, Stanford Online Apr 2024 – Jun 2024
Section Leader Remote
- Selected from over 2,000 applicants to teach the Stanford CS106A curriculum, enabling students from non-technical backgrounds to gain hands-on programming experience.
- Facilitated weekly sessions, providing personalized instruction and solidifying core programming concepts for a group of international learners.

Skills

Programming Languages — Python | Verilog | C | MATLAB

Frameworks & Libraries — TensorFlow | Transformers | NumPy | Pandas | Streamlit | OpenMP

Tools and Platforms — Vivado | VS Code | Git | GitHub | Linux | Google Colab